

ISAACK JOSHUA LUKUMAY

Machine Learning Engineer · Software Developer

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PROFILE

Machine learning engineer and software developer who builds end-to-end systems — not just models. Work spans ML/AI, backend engineering, cross-platform desktop applications and mobile, with a consistent focus on getting systems into real use: offline and on-device inference, explicit safety boundaries, and modular architecture that survives a second use case. Final-year Computer Science student at St. Joseph University in Tanzania and machine learning intern at the Emerging Technologies for Healthcare Lab, MUHAS.

EXPERIENCE

Machine Learning Intern

March 2025 – Present

Industrial Practical Training

Emerging Technologies for Healthcare (ETH) Lab, Muhimbili University of Health and Allied Sciences (MUHAS) — Dar es Salaam, Tanzania

- Designed and deployed **Mwana AI**, a breast cancer ultrasound classifier (PyTorch, ONNX, Flutter) built for offline use in clinics.
- Built a CNN-based TB/HIV co-infection detection system with Grad-CAM explainability.
- Led multi-institution data extraction for a respiratory disease study across Aga Khan, Warwick and NTLP.
- Assisted in fine-tuning the RETFound foundation model for diabetic retinopathy detection.
- Applied transfer learning to cardiac imaging for dilated cardiomyopathy prediction.
- Designed data preprocessing pipelines for medical imaging and audio datasets.

PyTorch · ONNX · Flutter · Grad-CAM · Transfer learning · Medical imaging

SELECTED PROJECTS

Lyceum — cross-platform desktop app that turns any LLM into a structured, interactive tutor. Vendor-agnostic: users bring their own key and choose Claude, GPT, Kimi or a local open-weight model.

Electron · TypeScript · React · Multi-LLM | github.com/Isaackjoshua/Lyceum

Triage — transport-agnostic AI agent that autonomously diagnoses faulty computers, applies safe software fixes, and escalates hardware or high-risk issues to a human operator.

Python · LLM tool-use · System diagnostics | github.com/Isaackjoshua/Triage

Afya-Predict — modular AI platform predicting disease outbreaks in Tanzania from health-facility, climate and mobility data, surfacing early-warning signals for public-health decision-makers.

Python · Machine Learning · Epidemiological modelling | github.com/Isaackjoshua/Afya_Predict

Amana — multisignature escrow platform securing mobile money for small merchants and gig workers across East Africa. Full transaction lifecycle in PostgreSQL with atomicity guarantees; OWASP-guided FastAPI service deployed behind nginx.

Python · FastAPI · PostgreSQL · Celery · Selcom API | github.com/Isaackjoshua/Escrow

Mwana AI — ultrasound image classifier for breast cancer screening, exported to ONNX for on-device inference with a Flutter Android front end, designed for low-resource clinical settings.

Python · PyTorch · HuggingFace · ONNX Runtime · Flutter | github.com/Isaackjoshua/breast_cancer_ai

TECHNICAL SKILLS

Languages & Tools	Python, TypeScript, Git/GitHub, Linux, nginx, Streamlit, Electron
Machine Learning	Deep learning, transfer learning, model fine-tuning, ONNX export, on-device inference, LLM tool-use, agentic systems
Frameworks	PyTorch, TensorFlow/Keras, HuggingFace Transformers, Scikit-learn, FastAPI, Celery, React
Computer Vision	Image preprocessing, augmentation, segmentation, classification, Grad-CAM explainability
Data	Data cleaning, EDA, feature engineering, PostgreSQL, data visualisation
Mobile & Desktop	Flutter (Android), Electron (cross-platform desktop)

EDUCATION

BSc (Hons) Computer Science	2023 – 2026
St. Joseph University in Tanzania — expected 2026	
ACSEE	2021 – 2023
Kongwa Secondary School, Dodoma	

LANGUAGES

English	Professional working proficiency
Swahili	Native

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